

Ananya Deshmukh

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Lead AI / ML Engineer (LLMs & RAG)

PROFESSIONAL SUMMARY

Applied Artificial Intelligence and Machine Learning Engineer with 4+ years of experience building and deploying production-grade LLM applications, RAG pipelines, and semantic document understanding systems. Deep expertise in Groq LLaMA, OpenAI APIs, LangChain, and Python vector search engines.

TECHNICAL SKILLS

Python, PyTorch, LangChain, LlamaIndex, Groq API (LLaMA 3), OpenAI GPT-4, Vector Databases (Weaviate, pgvector, Pinecone), FastAPI, Document OCR/Parsing, NLP Semantic Search, Docker, AWS Sagemaker.

PROFESSIONAL EXPERIENCE

Applied AI Lead Engineer | CognitiveAI India, Hyderabad (2022 - Present)

- Architected and deployed production RAG document analysis pipelines using Python, FastAPI, and Groq LLaMA-3 models.
- Built multi-format document parser (PDF, DOCX) with intelligent layout detection and semantic vector embeddings.
- Optimized LLM prompt guardrails and evaluation metrics, improving automated scoring accuracy by 38% over baseline keyword matching.
- Collaborated with product engineers to serve AI endpoints with sub-500ms response times under heavy load.

Machine Learning Engineer | DataSense Systems, Hyderabad (2020 - 2022)

- Developed NLP classification and entity extraction models using PyTorch and HuggingFace transformers.
- Deployed ML inference services in Docker containers on AWS ECS with asynchronous task queues.

EDUCATION & CERTIFICATIONS

M.Tech in Artificial Intelligence & ML | IIIT Hyderabad (2018 - 2020)